

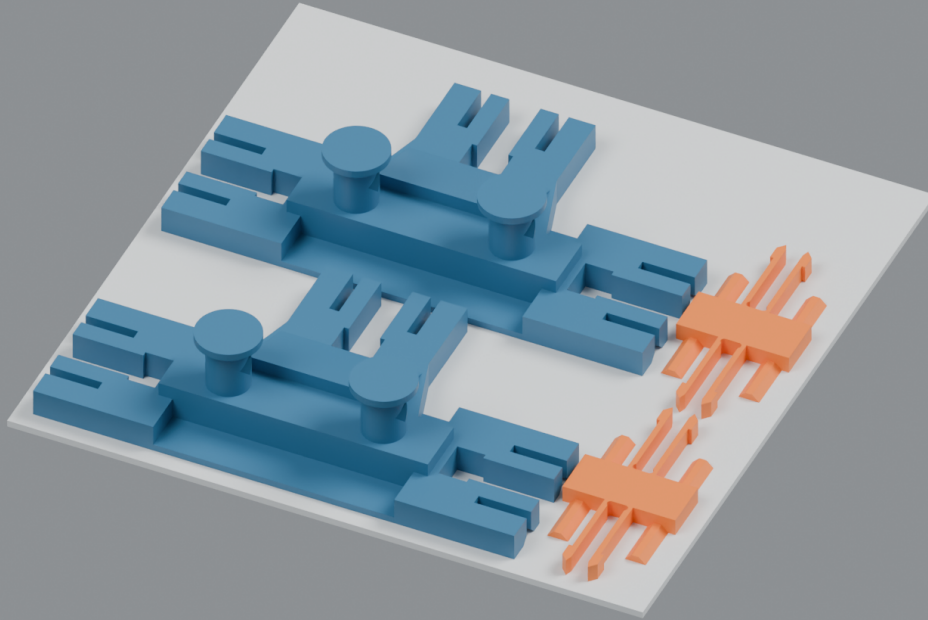
First frame print / A1 Mini

FIELDWORK / FPV RACING GATE

01

One plate / four useful test pieces

Two identical branch rails and two click keys. The keys print flat, rotated 90 degrees in the narrow strip beside the rails.



Choose FIRST_FRAME_TEST. All four pieces are production geometry; no magnets, PVC or additional supplies are needed for the click test.

Recommended: FIRST_FRAME_TEST_A1Mini_PETG.3mf

Two identical EDGE_BRANCH rails + two identical CLICK_KEYS.

80.14 g PETG / about 3 h 20 min / one plate / four pieces.

Optional alternative: FIRST_FRAME_TEST_WITH_PAD_A1Mini_PETG.3mf adds one PAPER_PAD. Five pieces / 81.63 g / 3 h 24 min. Print one file, not both.

A1 Mini, 0.4 mm nozzle, PETG, 0.20 mm layers, 4 walls, 20% infill, no supports/brim, print by layer. Open as a project; use 100% scale and the supplied flat orientation. Render colors are illustrative; one filament prints everything. Select your actual PETG preset and re-slice if it differs.

5 mm minimum bed margin; 29 mm maximum height. All meshes match the production parts. The second key is a spare/comparison. No extra hardware is needed for this click test.

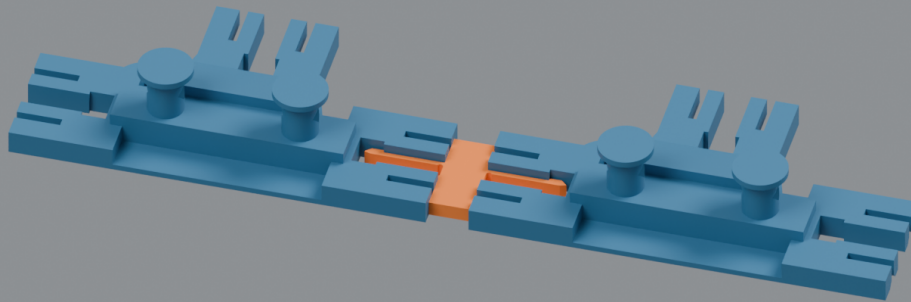
1 / Connect the rails end-to-end

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02

Test the straight rail connection

Use one key to join the two rails end-to-end. Keep their smooth paper faces coplanar and both side branches facing the same way.



Seat both rigid guide runners and inspect both catches at each end. The second key is a spare / comparison part.

Prepare

Cool and remove the print. Clear loose strings and first-layer burrs from the receiver and the two central spring prongs. Preserve the hook shoulders. Keep the spare key untouched for comparison.

Slide and latch

Keep smooth paper faces coplanar and both side branches facing the same direction. Insert the key along the rail axis. The wide outside runners guide it; the central fork flexes inward and springs outward behind the receiver shoulders. Check both hooks at each end. Support close to the joint when pressing.

Check

The straight assembly is 286.63 mm long. Apply a gentle axial hand pull and check for unintended release or obvious play. Check slight bending near the joint. Stop if cracking or whitening appears. Do not use the full rail length as a bending lever.

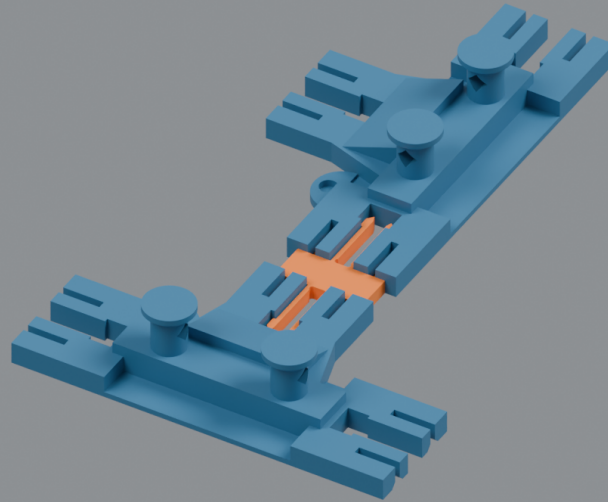
2 / Release, then build the T joint

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03

Release it / rebuild as a T joint

Move the same key into the projecting side socket. Connect an end of the second rail at 90 degrees; use the same flat orientation.



This tests the actual crossmember interface, even though both test rails have branches. Pinch the two spring tips to release each key end.

Release

Access the central fork from the open back. Press both hook tips inward toward the center and withdraw straight. Repeat at the other end. Do not twist or pry upward. If comfortable hand release is not possible, record the problem instead of forcing it.

Reconfigure

Move the same key into the projecting side receiver of one rail. Attach an end of the second rail at 90 degrees, with both paper faces in the same plane. This is the actual gate crossmember interface. The two configurations are tested successively, not assembled simultaneously.

Repeat

Perform 20 assembly/release cycles in each configuration, then compare the spare key. Record receiver tightness, release access, increasing play, whitening, cracking or permanent prong bending. Success means both hooks latch reliably and release by hand without damage. This is an initial screen, not a strength or fatigue rating.

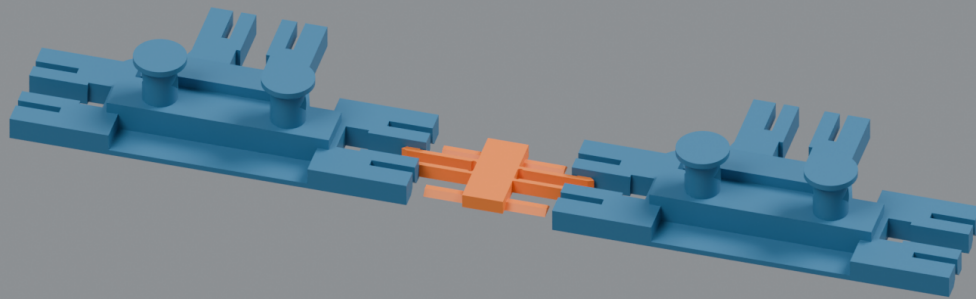
3 / Alignment and optional magnet test

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04

Slide the guides straight into the receiver

The wide outside runners guide the joint. The two narrow central prongs flex inward and their hooks catch the receiver shoulders.



Do not push the key down from above or twist it out. Pinch both accessible hook tips toward the center, then slide the joint apart.

Slide the guides along the rail axis; do not push the key down from above. Test results should identify whether an end receiver, side receiver or particular key needs adjustment. Keep all parts at 100% scale.

Paper pads are optional

For a local paper-grip comparison, add two 6 x 2 mm magnets and paper. Check polarity before fixing magnets with compatible adhesive. Without a pad: rear magnet / **unchanged 0.4 mm frame skin** / paper / bare front magnet. With a pad: rear magnet / frame skin / paper / 0.4 mm pad skin / front magnet. The pad offers a larger contact area and a handle; compare grip, marking and removal. Reuse the front magnet between tests before bonding, or use a third magnet.

Next stage

After click fit passes, test one RUNG_SLEEVE with actual PVC, then the 609.6 mm-wide braced bay in the full gate guide. The peaked sleeve roof is unchanged. These parts are reusable in the gate. This first plate tests joinery, not full-width rigidity or wind performance. No physical prototype has been tested here.